

David Kreuter

108 West Cherbourg Dr, Cheektowaga, NY | +1 716-444-9433 | davidckreuter@gmail.com

Senior Infrastructure & Networking Engineer | High-Performance Systems | Automation at Scale

Seasoned technical professional with 10+ years of experience in engineering high-performance infrastructure and networking solutions across enterprise environments. Proven expertise in network architecture, performance tuning, embedded systems troubleshooting, and large-scale automation--including building a global provisioning framework that manages 20,000+ virtual machines through integrated API-driven workflows. Possesses a deep background in customer-facing engineering, cross-functional collaboration, and mentoring technical teams. Skilled at diagnosing complex hardware and software interactions, from NUMA topology optimization to firewall policy automation, with a track record of independently driving projects from concept to delivery in fast-paced, high-stakes environments.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Architectural Scale:** Engineered a global server provisioning framework that deploys and manages 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate compute placement and backup orchestration, ensuring 100% RTO/RPO compliance and optimal resource allocation at build time.
- **Infrastructure Resilience:** Engineered and maintain a highly resilient infrastructure spanning three identical, geo-redundant datacenters, ensuring maximum uptime and disaster-recovery readiness--directly analogous to hyperscale AI factory reliability requirements.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation and a hardened security posture across complex, multi-zone network topologies.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application that serves as middleware between ServiceNow and the Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume, low-latency scaling.
- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines using Terraform to automate the deployment of virtual machines to Azure; implemented a pull-request-based workflow that standardizes infrastructure changes through automated plan/apply cycles.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premise automation workflows to be fully cloud-agnostic, enabling seamless deployment to Azure; designed abstraction logic that is transparent to end users, maintaining a consistent interface while handling underlying infrastructure complexity.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivers fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and optional hardened SQL deployments at scale.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code--driving cross-team technical alignment.
- **Operational Efficiency:** Developed an intelligent IP provisioning engine through InfoBlox API integrations, eliminating 1,000+ annual engineer hours and resolving a multi-year infrastructure bottleneck by applying deep networking protocol expertise.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premise SQL cluster, enabling a unified DBaaS offering that consolidates fragmented legacy databases and streamlines application delivery.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict three-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in 'Read-Only' dry-run modes; implemented Least Privilege permissions and robust rollback protocols for stale object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability--demonstrating deep hardware-level troubleshooting skills.
- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance across distributed infrastructure.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premise networks with cloud resources, ensuring seamless hybrid-cloud operations and low-latency connectivity.
- **Security Architecture:** Engineered a secure reverse proxy solution that protects sensitive internal infrastructure, enabling customer online ordering while mitigating external threat vectors.

Network Engineer

- **Network Administration:** Managed complex firewall rules and switch ACLs for a massive distributed network spanning 12

airports and three satellite offices, requiring deep knowledge of L2-L4 networking protocols.

- **Proactive Monitoring:** Developed service-aware alerting frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and performance analysis.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security posture.
- **Enterprise Mobility:** Onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geo-fencing, remote-wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring by designing custom Grafana dashboards to visualize real-time performance metrics across networked systems.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster-recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Networking: High Availability, BGP, DNS, VPN, VRRP, F5 Load Balancing, Intrusion Detection, Vulnerability Scanning, Meraki MDM

Infrastructure: NUMA Tuning, VMware vSphere, OS Hardening (CIS/STIG), Check Point Firewalls, Infrastructure as Code (IaC), Azure Cloud Services, Nutanix AHV, Rubrik, Active Directory, DBaaS

Languages: Python, Bash, PowerShell, SQL, YAML, Javascript, PHP

Automation/Orchestration: Ansible Automation Platform, REST API, CI/CD Pipelines, Terraform, GitOps, Middleware Development, Gitlab, Grafana, GraphQL, ServiceNow, Unit Testing

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tift Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present